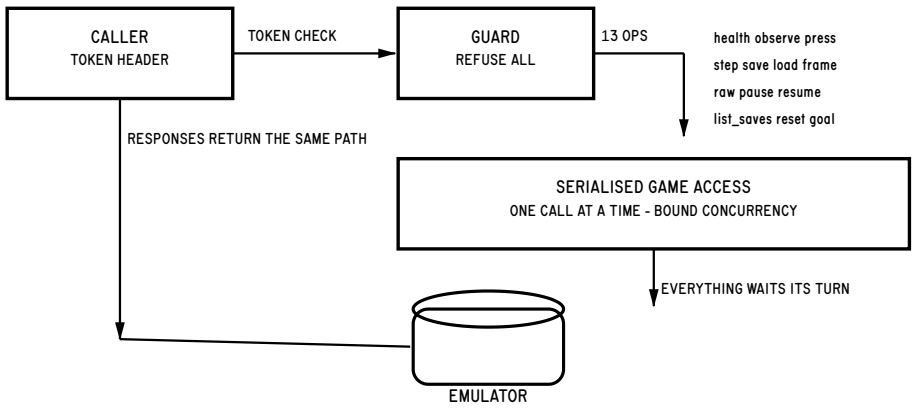
PART 2 · SHEET 2.1-1

EMULATOR BRIDGE

2.1.1 INTRODUCTION.

The bridge is the only path to the emulator. It binds loopback only, refuses every request that does not carry its token, and exposes thirteen operations. A session without the token cannot talk to the emulator at all (figure 2.1-1).



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Figure 2.1-1. Token Gate, Operation Dispatch, Serialised Game Access

2.1.2 FUNCTIONAL DESCRIPTION.

Every caller presents the token; the guard compares it in constant time and refuses the rest. Operations are line-delimited requests on one socket or single commands. Game access is serialised: one call at a time, so a console viewer and an agent cycle queue rather than corrupt the emulator.

2.1.2.1 Serialisation.

An unbounded thread per connection once drove the host to load 1016. The bound now caps concurrency and Game access is a lock, so one slow frame makes everything wait its turn rather than fight.

2.1.2.2 One Socket Or One Command.

A live session speaks line-delimited requests on one socket; everything else sends single commands. When one Game call stopped answering it held the bridge for 25 seconds and every panel starved; the bound now makes one slow frame cost itself only.

2.1.2.3 Revocation.

The bridge refuses to boot without a token file, so it cannot start naked by accident. Revoke by killing the process and deleting the token; enable per session.

2.1.3 PERFORMANCE AND DESIGN DATA.

Table 2.1-1. Bridge Data

Parameter	Value
Listen	127.0.0.1:8770, loopback only
Auth	token, compare_digest
Operations	13 (health ... goal)
Frame, native	160 x 144 px, about 2.7 kB
Frame when degraded	412 B blank - restart bridge
Step speed, measured	120 frames in 6 ms
Capacity used at 60 fps	about 0.3 percent

CAUTION

TWO DRIVERS ON ONE BRIDGE FIGHT. STOP THE CONSOLE LIVE TICKER BEFORE AN AGENT RUN.

BRG